Supplementary Material for STORM-CARE: A Physics-Informed Pretrained World Model for Disaster-State Forecasting and Counterfactual Intervention Simulation

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This supplement is self-contained: it restates the evaluation protocol where needed so that each section can be read independently of the main paper. Section S1 details data sources and the frozen storm-level split; Section S2 specifies the construction of the proxy-grounded humanitarian targets, which the main paper defers here; Section S3 lists architectures and hyperparameters; Section S4 gives complete physics-residual definitions, weights, and the full physics-weight sweep; Section S5 details the self-supervised pretraining protocol and extended RQ3 results; Section S6 reports the complete ERA5-complete Irma/Ian learned-model case study; Section S7 extends the humanitarian evaluation with unnormalized errors and baseline configurations; Section S8 details uncertainty-cone construction and full calibration results; Section S9 reports *all* intervention channels, including the weak and counterintuitive non-evacuation responses summarized only qualitatively in the main paper; Section S10 presents the Hurricane Ian qualitative case study; Section S11 provides the reproducibility checklist, compute resources, and artifact manifest; and Section S12 extends the limitations and societal-impact discussion.

S1 Data Sources, Preprocessing, and Split Composition

S1.1 Sources

- **HURDAT2** (Landsea and Franklin 2013): Atlantic best-track positions, maximum sustained wind, and central pressure at 6-hour intervals. We use the 2024 Atlantic HURDAT2 as released by NHC, with records through 2024-11-18; NHC HURDAT2 files carry no embedded revision-date field, so no exact download timestamp is recoverable from the file itself.
- **Synthetic storm augmentation**: foundation-model pretraining augments the HURDAT2 records with synthetically generated global storms produced by a seeded scenario generator. IBTrACS is *not* consumed by the frozen configuration (`ibtracs_path=None` in the saved checkpoint config); the pipeline falls back to HURDAT2 plus the synthetic generator, and no IBTrACS version claim applies to the reported runs.
- **ERA5** (Hersbach et al. 2020): reanalysis fields where available in the current repository. Variables retrieved: zonal wind u, meridional wind v, and geopotential z at pressure levels 850 hPa and 500 hPa, at $0.25^\circ \times 0.25^\circ$ resolution (ECMWF, retrieved via `cfgrib`). Test-split coverage: 1 of 107 test storms has any ERA5 file; dataset-wide, 86 of 11,087 observations (0.8%) have ERA5 coverage. This substantiates the main paper's statement that ERA5 inputs are unavailable for nearly all test storms.
- **CDC SVI** (Flanagan et al. 2011): 2022 release, Florida tract geography, four ranked themes (`RPL_THEME1`–`RPL_THEME4`) plus the composite `RPL_THEMES`. SVI-derived vulnerability features are consumed by the Module 1 foundation encoder only; they do not enter the Module 3 proxy-label generator (S2).
- **Facility and population layers**: schools, hospitals, shelters, and population clusters are *synthetic*, generated alongside the scenario data with synthetic identifiers; no external facility snapshot (e.g., HIFLD/OSM/HDX) is consumed. The processed facility table joined to SVI themes for Module 1 features contains 351 hospital, 1,978 shelter, and 5,381 school rows.

S1.2 Frozen storm-level split

The temporal partition, frozen before any test evaluation, contains 342 training storms (1995–2015), 70 validation storms (2016–2019), and 107 test storms (2020–2024), for 519 Atlantic tropical cyclones total (Table 1). No storm appears in more than one partition. Foundation-model pretraining additionally partitions its own records by natural storm identity with zero storm-identity overlap between its training and validation sets. The per-horizon forecast-window counts n for the test split are those of Table 1 of the main paper (2998, 2891, 2678, 2267, 1885, 1256 at 6–120 h); forecast-window counts for the train and validation partitions are not a committed artifact. Foundation pretraining uses 400 records in 325 storm-identity groups (split seed 42): 324 training records (260 groups, 837 windows) and 76 validation records (65 groups, 238 windows), with zero storm-ID and zero group-key overlap. The canonical split definition (`splits/storm_splits.json`) has SHA-256
`87bae628702eaf8bf887ba039b9f8aeff1e70123099c5b5a33b9d6a338f27959`.

Partition	Years	Storms
Train	1995–2015	342
Validation	2016–2019	70
Test	2020–2024	107

Table 1: Frozen storm-level split composition. Test-split per-horizon window counts appear in main-paper Table 1.

S1.3 Preprocessing and feature construction

Best-track records are aligned to the native 6-hour HURDAT2 grid; ERA5 fields, where present, are aligned to the same grid, and missing ERA5 fields are recorded as unavailable rather than zero-imputed, matching the evaluation-scope statement in the main paper. Normalization statistics are computed on training storms only. Atmospheric node features comprise wind components, pressure, and available reanalysis variables; region and population node features combine SVI-derived vulnerability features (consumed by Module 1) with synthetic population-cluster features; facility node service and capacity attributes are synthetic (S1.1, S2).

S2 Construction of Proxy-Grounded Humanitarian Targets

The main paper states that humanitarian targets are simulator-derived proxies and defers their construction here. This section specifies the generative protocol so that the scope of the RQ1b claims is auditable.

S2.1 Design intent

Observed event-level labels for population exposure, school disruption, hospital accessibility, shelter demand, and recovery priority are not consistently available at the spatial and temporal granularity of the disaster graph. The proxy protocol therefore tests whether the model learns *hazard-to-impact relationships under a specified generative process*, not whether it predicts realized post-disaster damage. All comparisons in Table 2 of the main paper (RF, MLP, XGBoost, STORM-CARE) are trained and evaluated under this same protocol, so the comparison is internally matched even though the targets are synthetic.

S2.2 Generative specification

Scenarios are produced by a seeded synthetic generator (`np.random.default_rng`) that instantiates a Rankine-vortex storm hazard over a disaster graph with, per scenario, 5 population clusters, 6 schools, 2 hospitals, and 3 shelters (the frozen checkpoint's configuration). The generator emits per-node damage scores $d_{(\cdot)} \in \mathbb{R}$, from which the five targets are computed deterministically; all stochasticity enters through the scenario seed, and no additive label noise is applied. Writing $\mathrm{clamp}(x) = \min(\max(x, 0), 1)$:

- **Exposed child count** (count): $y_{\mathrm{exp}} = w_{\mathrm{pop}} \cdot f_{\mathrm{child}} \cdot \mathrm{clamp}(d_{\mathrm{pop}}) \cdot 20{,}000$, where w_{pop} is a normalized population-cluster weight, f_{child} is the cluster child fraction, and 20,000 is a fixed per-cluster population scale constant. This constant is *not* derived from any real population source; no real population, SVI, or facility data enters the label generator.

- **School disruption** (binary): $y_{\mathrm{sch}} = \mathbf{1}[\,\mathrm{clamp}(d_{\mathrm{sch}}) > 0.15\,]$, with disruption threshold 0.15.
- **Hospital accessibility** (unitless $[0,1]$ index): $y_{\mathrm{hos}} = \mathrm{clamp}(1 - d_{\mathrm{hos}})$.
- **Shelter demand** ($[0,1]$): $y_{\mathrm{shl}} = \mathrm{clamp}(0.55\, d_{\mathrm{shl}} + 0.45\, c_{\mathrm{shl}})$, where c_{shl} is the synthetic shelter capacity feature.
- **Recovery priority** (ranking score in $[0,1]$): $y_{\mathrm{rec}} = \mathrm{clamp}(d_{\mathrm{reg}})$, computed from regional infrastructure damage.

S2.3 Scenario separation

Training and held-out evaluation scenarios use disjoint random seeds: scenario generation uses training seed 123 and held-out evaluation seed 999, for both the graph model and all tabular baselines. No scenario parameters were adjusted after observing held-out results.

S3 Architectures and Hyperparameters

Table 2 lists the configuration of each module, taken from the `cfg` dictionaries embedded in the saved (frozen) checkpoints rather than from class defaults. All modules train for 20 epochs with AdamW (weight decay 10^{-4}). Training wall-clock per module is reported in S11.2; recorded random seeds are 42 (world model and tabular baselines) and 123/999 (humanitarian scenario training/held-out evaluation).

Module	Component	Hyperparameter	Value
M1	Transformer encoder	layers / heads / width	3 / 4 / d_{model}=128, d_{ff}=512
		optimizer / LR / schedule	AdamW / 2×10^{-4} / LambdaLR, linear warm-up 5 of 20 epochs, then decay
		batch size	16
		pretraining epochs; selected epoch	20; 20
M2	Graph–Fourier operator	message-passing (GNO) layers	2
		Fourier (FNO) layers	2
		Fourier modes / width	6×6 / 32 (d_v=32, d_{hidden}=64)
		optimizer / LR / schedule	AdamW / 10^{-3} / LambdaLR
		γ default	1.0
M3	Heterogeneous GNN	layers / node types / edge types	2 / 6 / 4
		widths	d_{hidden}=32, d_{type}=8, d_{edge}=8
		head architectures (5 targets)	shared shape: Linear$(d, d/2)$ $\to$ GELU $\to$ Linear$(d/2, 1)$ $\to$ Sigmoid; the hospital head outputs $1 - \sigma(\cdot)$
		state head (M4 handoff)	2-layer MLP, no final activation, d=32
		optimizer / LR / schedule	AdamW / 10^{-3} / CosineAnnealingLR ($T_{\max}$= epochs, $\eta_{\min}$=0.01 LR)
M4	RSSM	d_z	16 (d_{hidden}=d_{enc}=d_{dec}=32)
		KL weight; prediction weight	β_{KL}=0.1; β_{pred}=0.5
		warm-up steps; rollout horizon	8; 8
		optimizer / LR / schedule	AdamW / 10^{-3} / CosineAnnealingLR
M5	Scenario engine	(no trained parameters)	—

Table 2: Module configurations, from the frozen checkpoints' embedded configs. M5 introduces no gradient-trained parameters; it branches held-out warm-up trajectories under the frozen M4 checkpoint.

S4 Physics Residuals: Definitions, Units, and Full Sweep

S4.1 Residual definitions

With $\nabla_{\mathcal{G}}$ and $\Delta_{\mathcal{G}}$ the graph finite difference and graph Laplacian computed on physical edge distances, and D_t the 6-hour temporal difference operator, the six residuals are

$$R_{\text{adv}} = \|D_t q + v \cdot \nabla_{\mathcal{G}} q\|_2^2, \qquad R_{\text{diff}} = \|D_t u - \kappa \Delta_{\mathcal{G}} u\|_2^2, \tag{1}$$

$$R_{\text{mass}} = \|D_t \rho + \nabla_{\mathcal{G}} \cdot (\rho v)\|_2^2, \qquad R_{\text{wp}} = \|B(v, p)\|_2^2, \tag{2}$$

$$R_{\text{cont}} = \|S^{-1}(\hat{u}_{t+1} - u_t)\|_2^2, \qquad R_{\text{nrg}} = \|E_{t+1} - E_t - W_t\|_2^2, \tag{3}$$

with default weights $(\beta_{\text{adv}}, \beta_{\text{diff}}, \beta_{\text{mass}}, \beta_{\text{wp}}, \beta_{\text{cont}}, \beta_{\text{nrg}}) = (1000, 500, 1000, 500, 1000, 100)$ and global multiplier γ. Pressure-gradient calculations use SI units and spatial derivatives are normalized by physical edge distances.

Implemented forms. The atmospheric state comprises seven channels $(u_{850}, v_{850}, u_{500}, v_{500}, z_{500}, T_{2\text{m}}, p_{\text{MSL}})$. The advected and diffused scalar is temperature ($q \equiv T_{2\text{m}}$), with diffusivity $\kappa = 10^3\,\text{m}^2\,\text{s}^{-1}$. Air density is the constant $\rho = 1.225\,\text{kg}\,\text{m}^{-3}$ (there is no density state channel), so the mass term reduces to a penalty on the divergence of the wind field. The wind–pressure residual is the gradient-wind imbalance

$$B(v, p) = \frac{V^2}{r} + fV + \frac{1}{\rho}\frac{\partial p}{\partial r}, \qquad V = \|v\|, \tag{4}$$

evaluated on the *predicted* next state, with the radial pressure gradient computed by a radial-gradient operator and r the physical radius from the domain center in meters (domain radius in degrees $\times$ 111,000). There is no single learned or named scale matrix S in the implementation: normalization is per-residual, using an acceleration scale for the advection and wind–pressure terms, a tendency scale for diffusion, and, for temporal continuity, an explicit per-channel scale vector $[\, s_{\text{wind}} \times 4,\ s_{\text{geopot}},\ s_{\text{temp}},\ s_{\text{pres}} \,]$ over the seven channels. In R_{nrg}, E is the kinetic energy computed from the wind channels (the physics audit reports this residual as the kinetic-energy term).

S4.2 Per-residual reductions at the default weight

At the default weight, the full-physics model reduces all six final validation residuals relative to the matched no-physics model: 78.0% (diffusion), 50.4% (wind–pressure consistency), 30.4% (mass consistency), 17.0% (temporal continuity), 1.3% (kinetic energy), and 0.9% (advection). Table 3 gives the absolute final validation values (track RMSE 0.022002 full physics vs. 0.019341 no physics), and Figure 1 shows the corresponding training-time residual trajectories.

Residual	Full physics (γ=1)	No physics	Reduction
R_{adv}	4.682×10^{-5}	4.725×10^{-5}	0.9%
R_{diff}	3.206×10^{-3}	1.458×10^{-2}	78.0%
R_{mass}	4.123×10^{-5}	5.928×10^{-5}	30.4%
R_{wp}	1.133×10^{-4}	2.286×10^{-4}	50.4%
R_{cont}	3.646×10^{-3}	4.395×10^{-3}	17.0%
R_{nrg}	7.373×10^{-3}	7.469×10^{-3}	1.3%

Table 3: Final validation residuals for the default-weight full-physics model versus the matched no-physics ablation.

S4.3 Full physics-weight sweep

Table 4 reports the sweep over $\gamma \in \{0.1, 0.3, 1.0, 3.0\}$ with an independently rerun no-physics control. Values already reported in the main paper: no-physics control track RMSE 0.01932 (and 0.01934 for the original matched ablation); default (γ=1.0) RMSE 0.02200; γ=0.1 RMSE 0.01894 with diffusion residual 0.00854 (vs. 0.01457 no-physics) and temporal-continuity residual 0.00387 (vs. 0.00440).

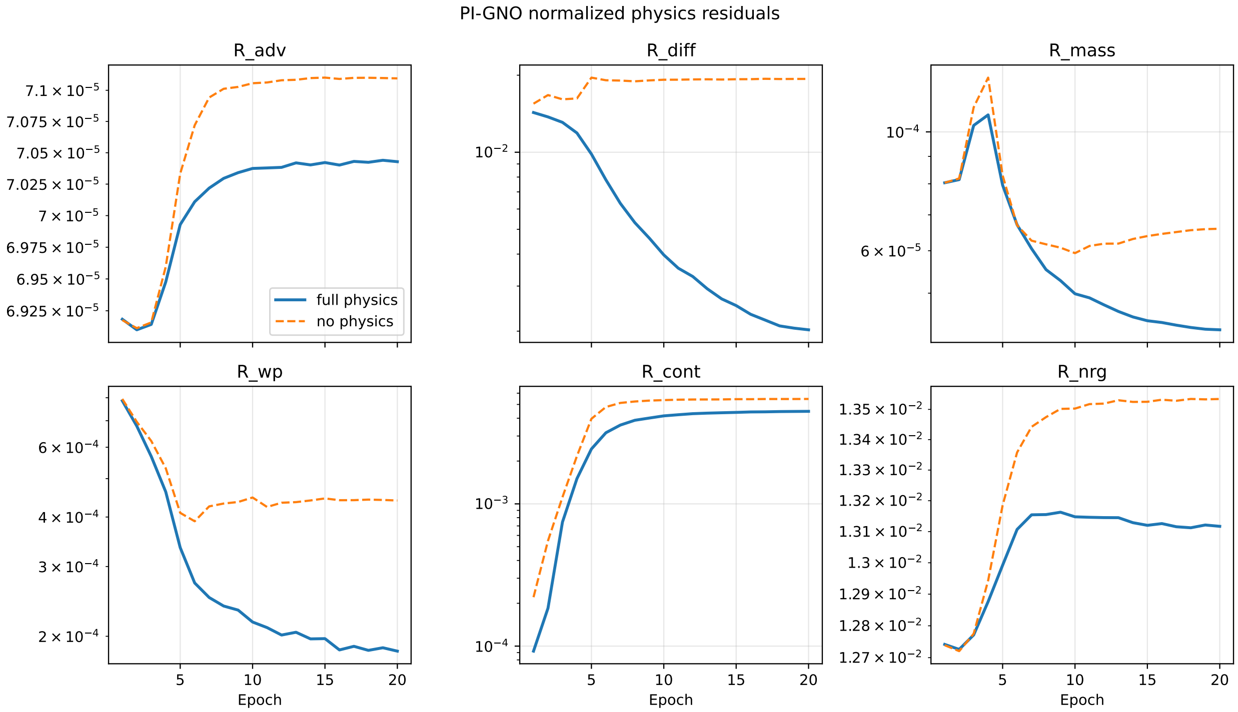

Figure 1: Normalized physics residuals over training for the default-weight full-physics model versus the matched no-physics ablation, from the committed training logs. These are training-set trajectories; the final *validation* values reported in Table 3 differ from the epoch-20 training-time endpoints shown here. The full-physics model attains lower residuals on all six terms. The upward drift of R_{cont} and R_{nrg} in both models reflects departure from near-trivial early-epoch predictions, so the meaningful comparison is between the two curves at matched epochs, not against epoch 1.

S5 Self-Supervised Pretraining: Protocol and Extended RQ3

S5.1 Objectives and weighting

The pretraining objective is

$$\mathcal{L}_{\mathrm{SSL}} = \lambda_{\mathrm{mask}}\mathcal{L}_{\mathrm{mask}} + \lambda_{\mathrm{fut}}\mathcal{L}_{\mathrm{fut}} + \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}} + \sum_{\Delta} \lambda_{\Delta}\mathcal{L}_{\mathrm{mh}}^{\Delta}, \qquad \Delta \in \{6, 12, 24, 48, 72, 120\}\,\mathrm{h}.$$

Pretraining runs 20 epochs with a 5-epoch linear warm-up (S3).

Checkpoints are selected by validation multi-horizon track error, computed as the mean over the six horizons; the selected checkpoint is epoch 20 (the final epoch of the 20-epoch schedule) with selection score 860.196 km, improving monotonically from 1018.6 km at epoch 1. All calibration tables and figures in the main paper derive from this single checkpoint.

S5.2 Extended pretraining-vs-random comparison

Main-paper values, matched validation windows: mean 6–48 h track error 449.7 km (pretrained) vs. 545.6 km (random initialization); future-intensity MAE 3.36 kt vs. 40.21 kt; mean absolute P90 coverage error 0.022 vs. 0.515. The comparison uses 238 validation windows with zero group overlap between pretraining and validation storms. Per-horizon track error for the selected pretrained checkpoint is 106.9, 231.0, 509.3, 954.1, 1357.8, and 2002.0 km at 6, 12, 24, 48, 72, and 120 h (the aggregate 449.7 km is the evaluator's window-level 6–48 h mean; the unweighted mean of the four per-horizon values is 450.3 km). Additional pretraining diagnostics at the selected epoch: linear-probe accuracy 0.896, masked-reconstruction MSE 0.036, contrastive alignment 0.983.

Config	Track RMSE	R_{diff}	R_{cont}
No physics (rerun)	0.01932	0.01457	0.00440
$\gamma = 0.1$	0.01894	0.00854	0.00387
$\gamma = 0.3$	0.02108	0.00418	0.00408
$\gamma = 1.0$ (sweep rerun of default)	0.02201	0.00321	0.00364
$\gamma = 3.0$	0.02642	0.00339	0.00266

Table 4: Physics-weight sweep (single training seed, 42; sensitivity evidence, not a fully powered hyperparameter study). Validation track RMSE in normalized units; the sweep decision artifact records track RMSE and the diffusion and temporal-continuity residuals per configuration. The γ=1.0 row is the sweep's independent rerun of the shipped default (0.02201 vs. 0.02200 for the original default run). Track RMSE degrades monotonically above scale 0.1. The logged CFL number exceeds 1 at every scale tested; this is a residual-evaluation limitation of the demo configuration, disclosed per the physics audit, and the sweep does not change that finding.

S5.3 Graph-component ablations

The main paper reports the qualitative pattern: removing storm-propagation edges worsens exposed-population error but improves recovery-priority correlation, whereas removing transportation edges shows the reverse. Table 5 quantifies this from the regenerated ablation artifact. The ablation runs retrain the graph model with the corrected multitask loss on 50 training scenarios (seed 123) for 20 epochs and evaluate on 50 held-out scenarios (seed 999); the ablation evaluator reports exposed-children *peak MAPE* and the recovery-priority Spearman correlation. Because the ablation full-graph model is retrained under this protocol, its row is internally comparable to the edge-removal rows but is *not* numerically comparable to the main-paper Table 2 checkpoint (which reports sMAPE under a separate evaluation); no dependency-edge ablation was run, matching the main paper, which claims only the propagation and transportation patterns. Both qualitative claims are reproduced: removing propagation edges worsens exposure error and improves the priority correlation, and removing transportation edges shows the reverse.

Variant	Peak MAPE (%) ↓	School AUC ↑	Hospital MAE ↓	Priority ρ ↑
Full graph (ablation protocol)	533.63	0.803	0.0185	0.275
− propagation edges	593.69	0.798	0.0214	0.378
− transportation edges	526.93	0.793	0.0189	0.138

Table 5: Graph edge-type ablations, retrained and evaluated under a matched protocol (train seed 123, held-out test seed 999). Peak MAPE is retained here because it is the metric emitted by the ablation evaluator; it is dominated by near-zero true-count denominators (S7.1) and is used only for within-protocol comparison, not as a headline metric.

S6 ERA5-Complete Learned-Model Case Study (Irma/Ian)

Because ERA5 inputs are unavailable for nearly all test storms in the current repository, learned-model comparisons are restricted to common ERA5-complete forecast windows from Hurricanes Irma and Ian with matched normalized inputs. This case study compares learned architectures under identical inputs; it is *not* independent-storm generalization, and the main paper draws no storm-level neural-superiority claim from it.

Table 6 reports the complete results. They reproduce every qualitative ordering stated in the main paper: Persistence is the lowest-error method at every horizon and averaged over 6–48 h; GNO+DynGNN outperforms Transformer at all four horizons and LSTM at 12, 24, and 48 h, and loses to LSTM at 6 h. Confidence intervals are omitted rather than computed over this small window set.

Baseline configurations (matched capacity). LSTM: a 1-layer `nn.LSTM` (input 32, hidden 64) with an operator encoder (width 32, output 96) for ERA5 inputs. Transformer: d_{model}=64, 4 heads, 2 layers, feed-forward width 128, dropout 0.1. DCRNN: DCGRU cell with hidden width 8 and k=2 diffusion hops, bidirectional diffusion convolution over an 8-neighbor k-NN grid graph. GNO+DynGNN: operator encoder (width 48, output 128) with a dynamic GNN

Method	6 h	12 h	24 h	48 h	Mean 6–48 h
Persistence	29.247	72.842	197.041	527.727	206.714
LSTM	83.445	286.149	442.575	873.994	421.541
Transformer	170.140	211.689	381.798	611.512	343.785
DCRNN	89.515	269.963	335.053	736.555	357.771
GNO+DynGNN	119.690	156.353	322.895	598.080	299.255

Table 6: Track error (km) on common ERA5-complete Irma/Ian forecast windows with matched normalized inputs (window-level case study; not held-out-storm generalization). Evaluation covers 9 held-out test windows; the 45 common ERA5-complete windows are split deterministically at the window level into 27 train / 9 validation / 9 test, learned checkpoints are selected by validation mean track error, and results are reported on the held-out test windows only. Confidence intervals are omitted given the small number of windows.

(node dimension 32, hidden 64, 2 layers). On the same windows, landfall-timing error is 2.4 h for Persistence and 1.0–1.2 h for the learned models.

S7 Extended Humanitarian Evaluation

S7.1 Unnormalized exposure error

Because sMAPE compresses scale information, the main paper additionally reports exposed-child-count MAE: STORM-CARE 298.5 vs. RF 267.6, XGBoost 263.5, and MLP 290.1. Ordinary MAPE is retained in the artifacts as a reference row only: STORM-CARE 469.4% vs. RF 721.5%, XGBoost 779.4%, and MLP 1070.2%. Note that MAPE *reverses* the sMAPE/MAE ranking — it favors STORM-CARE only because it penalizes baseline under-prediction disproportionately near zero true counts — which is precisely why sMAPE and MAE are the headline metrics; we report the reversal rather than selecting the favorable metric. Per-scenario error distributions were not retained by the evaluation pipeline; only aggregate metrics per model are reported.

S7.2 Baseline configurations

The tabular baselines use fixed configurations set a priori (not tuned on held-out scenarios): random forest with 50 trees (`RandomForestRegressor`, `random_state=42`, library defaults otherwise); XGBoost with 100 trees, maximum depth 4, learning rate 0.1 (`random_state=42`); and an MLP with hidden layers $(32, 16)$ and at most 200 iterations (`random_state=42`). All baselines receive the same label-free 63-dimensional feature vector: mean, standard deviation, and maximum of the 7 raw node features (21 dimensions) plus per-node-type means over the 6 node types (42 dimensions), establishing input parity with the graph model at the summary-statistic level.

S7.3 Metric definitions

sMAPE is used as the headline exposure metric because ordinary MAPE is unstable when true exposed-population counts approach zero. The implemented variant includes the factor of two in the numerator,

$$\text{sMAPE} = \frac{100\%}{n} \sum_{i=1}^{n} \frac{2\,|\hat{y}_i - y_i|}{|y_i| + |\hat{y}_i|},$$

bounded on $[0, 200]\%$ with a symmetric denominator. School-disruption AUC is the standard ROC AUC computed on the binary disruption labels pooled over all held-out school nodes. Recovery-priority ρ is the Spearman rank correlation between predicted and target priority scores over held-out region nodes.

S8 Uncertainty-Cone Construction and Full Calibration

S8.1 Cone construction

The cone is parametric: an axis-aligned Gaussian ellipse with independent per-axis standard deviations and no cross-covariance term, a position (ϕ, λ) falling inside the level-z cone iff

$$\left(\frac{\phi - \mu_\phi}{\sigma_\phi}\right)^2 + \left(\frac{\lambda - \mu_\lambda}{\sigma_\lambda}\right)^2 \leq z^2, \tag{5}$$

with per-horizon parameters and z set from two-dimensional χ^2 cutoffs: $z_{\mathrm{P50}} = 1.177$ and $z_{\mathrm{P90}} = 2.146$. The same convention is used in the track pipeline and the foundation evaluation. No validation-time or post-hoc recalibration is applied to any reported number, per the main paper.

S8.2 Full coverage results

Empirical P90 coverage for the epoch-20 checkpoint: 0.891 (6 h), 0.913 (12 h), 0.886 (24 h), 0.845 (48 h), 0.716 (72 h), 0.487 (120 h), against the nominal 0.90 target. The corresponding P50 coverages are 0.588, 0.576, 0.489, 0.428, 0.331, and 0.210 against the nominal 0.50 target. Table 7 reports the effective per-horizon calibration counts. The model is approximately calibrated at short horizons and increasingly overconfident with horizon; the degradation is reported raw rather than recalibrated post hoc.

Horizon	6 h	12 h	24 h	48 h	72 h	120 h
n_{valid}	238	231	219	194	169	119

Table 7: Effective per-horizon calibration counts for the epoch-20 foundation checkpoint (validation protocol).

S9 Complete Intervention Results, Including Failure Cases

The main paper reports the evacuation dose–response as the supported counterfactual finding and summarizes the remaining channels qualitatively as weak or counterintuitive. This section reports *all* channels so the limitation is auditable rather than asserted.

S9.1 Protocol

All interventions follow the warm-up-intervention protocol: intervene on the held-out warm-up state via $\tilde{o}^{(a)}_{t-C:t} = \mathcal{I}_a(o_{t-C:t})$, re-encode through the RSSM posterior, roll forward exclusively through the learned prior. No future decoded outputs are edited. Evaluation uses all 24 held-out sequences available to the loaded demo world-model checkpoint. Headline results use a 12-step rollout horizon with 5 Monte Carlo samples per rollout; the sensitivity check increases sampling to 60 (a 12× increase). The loaded demo world-model checkpoint is trained on 120 synthetic sequences. For each scenario, the artifacts record five outcome statistics averaged over the 24 sequences: peak exposure, shelter shortfall, final infrastructure damage, resource deficit, and mean hazard.

S9.2 Evacuation dose–response (supported finding)

Reproduced from main-paper Table 3: mean peak exposure with bootstrap 95% CIs over the 24 sequences — baseline 0.2915 [0.2842, 0.2990]; -12 h 0.2831 [0.2754, 0.2910]; -24 h 0.2787 [0.2715, 0.2864]; -36 h 0.2754 [0.2689, 0.2824]. The complete monotone ordering is reproduced in all 10,000 sequence-level bootstrap resamples, $P(\text{monotone}) = 1.000$.

As an additional direction check, *delayed* evacuation raises mean peak exposure to 0.2969 ($+1.85\%$ vs. baseline), completing the ordering $-36\,\mathrm{h} < -24\,\mathrm{h} < -12\,\mathrm{h} <$ baseline $<$ delayed.

Table 8 reports the one-sided paired Wilcoxon tests for the three adjacent comparisons with explicit Holm-adjusted p-values, computed by the same step-down procedure used for all significance reporting in the evaluation pipeline. All three comparisons remain claimable at $\alpha = 0.05$ after correction; the three adjusted values coincide because Holm's step-down monotonicity constraint propagates the largest rank-scaled value ($3 \times 9.08 \times 10^{-6}$) across the family.

Adjacent comparison	Mean difference	p (raw, one-sided)	p (Holm)
-12 h vs. baseline	-0.00843	9.08×10^{-6}	2.72×10^{-5}
-24 h vs. -12 h	-0.00438	1.03×10^{-5}	2.72×10^{-5}
-36 h vs. -24 h	-0.00333	1.92×10^{-5}	2.72×10^{-5}

Table 8: Evacuation dose–response: one-sided paired Wilcoxon tests over 24 held-out sequences with Holm step-down correction (3-test family). All three adjacent-step reductions remain significant at $\alpha = 0.05$ after correction.

S9.3 Non-evacuation channels (reported limitations)

Table 9 reports mean peak exposure over the 24 held-out sequences for every non-evacuation intervention (baseline 0.2915). Expected signs refer to the direction of change in peak exposure a domain reading would predict.

Channel	Intervention	Expected	Peak exp.	Δ vs. base	Assessment
Facility	Shelter capacity failure	$+$	0.2842	-2.50%	counterintuitive sign
Facility	Hospital service failure	$+$	0.2843	-2.47%	counterintuitive sign
Route	Road blockage	$+$	0.2902	-0.45%	near-null, wrong sign
Intensity	Storm intensity increase	$+$	0.2970	$+1.89\%$	correct sign, weak
Intensity	Storm intensity decrease	$-$	0.2854	-2.09%	correct sign, weak
Resource	Additional emerg. resources	$-$	0.2916	$+0.03\%$	null effect

Table 9: Non-evacuation intervention channels under the demo-scale checkpoint: mean peak exposure over 24 held-out sequences and percent change versus the factual baseline (0.2915). "Expected" is stated for interpretation only; no intervention is trained or selected to match it. Facility-failure and route interventions show wrong-sign or near-null responses, intensity responses carry the correct sign but small magnitude relative to the imposed hazard change, and the resource intervention is null — reported as limitations, matching the main paper.

Increasing Monte Carlo sampling from 5 to 60 samples per rollout does not change the sign or magnitude of these responses — e.g., the shelter-failure peak-exposure delta is -0.0073 at 5 samples and -0.0074 at 60 — indicating they arise from the learned checkpoint rather than sampling noise. Mirror diagnostics further confirm the outcomes are not numerical echoes of the intervention inputs: for example, the earlier-evacuation intervention applies a -0.12 input change while the observed peak-exposure delta is -0.0084, and no scenario's observed delta mirrors its input magnitude. The likely cause of the wrong-sign facility and route responses is a spurious infrastructure-to-exposure correlation learned from the small synthetic training set; correcting it would require longer or larger-scale world-model training, which is out of scope for the demo checkpoint and is disclosed as a limitation.

S10 Hurricane Ian Qualitative Case Study

Hurricane Ian (AL092022) illustrates the end-to-end pipeline. The underlying benchmark contains four Ian forecast windows and 20 model-prediction rows. The humanitarian and intervention maps are proxy/synthetic research outputs, not observed Hurricane Ian damage, and true geographic coordinates are used for all map panels.

Figure 2 shows the regenerated publication multipanel (track map with best track, uncertainty cones, trajectory errors, and the humanitarian-impact and evacuation-intervention views). Axes use true latitude and longitude (western longitudes negative, not inverted); coastlines are schematic offline context layers, not operational NOAA GIS products.

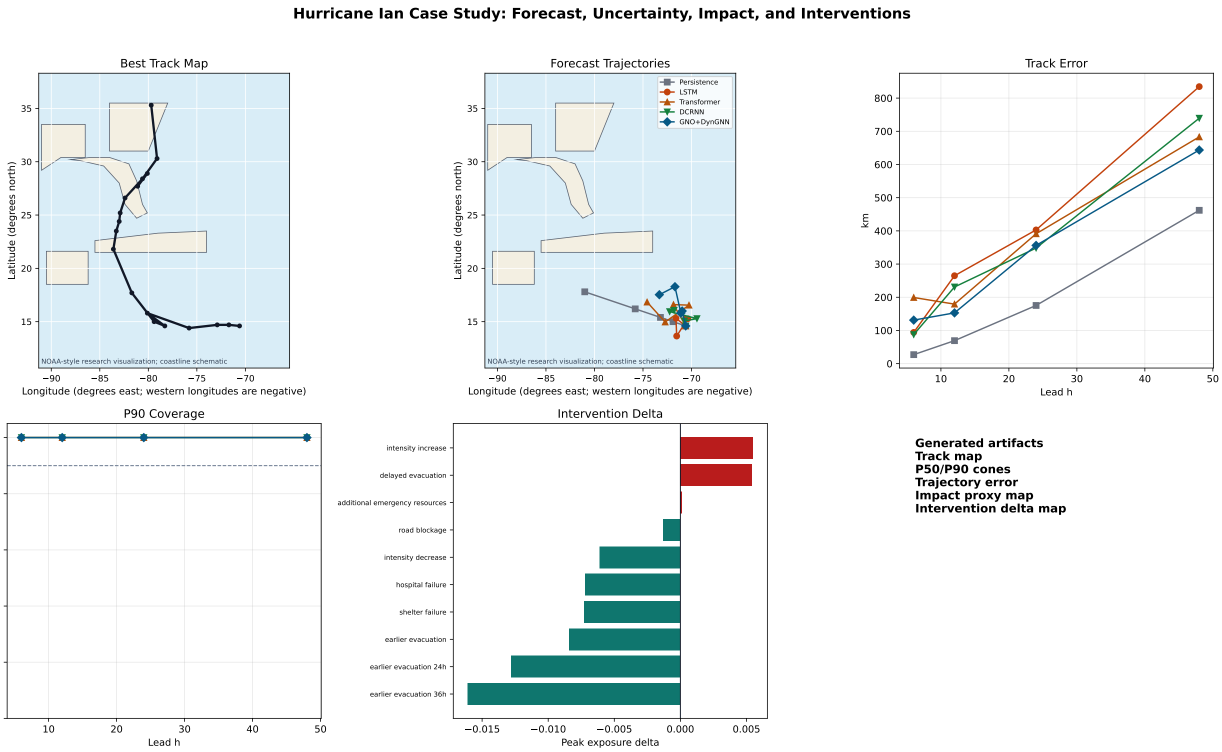

Figure 2: Hurricane Ian (AL092022) end-to-end case study: forecast track versus best track with P50/P90 uncertainty cones, per-lead trajectory errors, and predicted humanitarian-impact and evacuation-intervention maps. The humanitarian and intervention panels are proxy/synthetic research outputs under the S2 generative protocol, *not* observed Hurricane Ian damage.

S11 Reproducibility Checklist, Compute, and Artifacts

S11.1 Reproducibility

- Splits are frozen before test evaluation; no storm appears in multiple partitions; pretraining validation is storm-identity-isolated. The frozen split definition (`splits/storm_splits.json`) has SHA-256 `87bae628702eaf8bf887ba039b9f8aeff1e70123099c5b5a33b9d6a338f27959`.
- Forecasts at time t use no observations later than t (open-loop).
- Uncertainty intervals resample at the storm level (test benchmark) or sequence level (world-model experiments), never treating correlated windows as independent.
- Checkpoint selection uses validation criteria only (epoch 20 by validation mean track error); no post-hoc reselection on test diagnostics.
- The physics-weight sweep uses a single training seed and is labeled sensitivity evidence.
- All tables and figures are generated from committed evaluation artifacts; Table 10 lists SHA-256 hashes for the key frozen artifacts.
- Code and evaluation artifacts will be released upon acceptance; the artifact hashes in Table 10 allow verification of the released materials against the frozen runs reported here.

Artifact	SHA-256 (first 16 hex)
splits/storm_splits.json	87bae628702eaf8b
tables/table1_track_error_vs_baselines.csv	1cd94f74f17e40a8
tables/table_case_study_track_error.csv	5c3688715e57e35d
tables/table_foundation_model_training.csv	dcd4544b717407e4
tables/table_calibration_cone_coverage.csv	a8c76cb238e746df
tables/table2_humanitarian_impact.csv	8f076bb7150a67f7
tables/table3_ablations.csv	5b3220c41e5f6f17
tables/table_counterfactual_outcomes.csv	c3a5dfd385223083
metrics/inference_test_predictions_all_models.csv	9c3a19647869ae03
metrics/foundation/foundation_eval_metrics.csv	d5a0ba773f9ad602
metrics/humanitarian/humanitarian_eval_metrics.csv	7f692ef5d8886a5a
metrics/counterfactual/counterfactual_outcomes.csv	58d52ecb1fcf69a5
metrics/ablations/foundation_ablation_metrics.csv	bd9753a407131807
metrics/ablations/graph_ablation_metrics.csv	031f7b6eeec3638a

Table 10: Artifact manifest (truncated hashes; full SHA-256 values are recorded in the repository's experiment log and validation report).

S11.2 Compute

All modules were trained CPU-only; no GPU was used at any stage (`torch.cuda.is_available()` is `False` in the training environment), so total accelerator-hours are zero. Measured wall-clock for the frozen runs (single CPU machine, ~2 threads): Module 1 (foundation pretraining, 20 epochs) 26.6 min real time (55.9 min user time); Module 2 (physics operator, 20 epochs) ~55 s; Module 3 (disaster graph, 20 epochs) ~34 s; Module 4 (world model, 20 epochs) 14.2 s real time. The Module 4 timing was obtained by rerunning the frozen configuration at its recorded seed (42), which reproduced the committed checkpoint and training log byte-for-byte; the live checkpoint was left untouched. The term "demo-scale" in the main paper corresponds to exactly these budgets.

S12 Extended Limitations and Societal Impact

Proxy supervision. Humanitarian results certify learned hazard-to-impact structure under a specified generative protocol (S2), not predictive skill on realized damage. Transfer from proxy to observed outcomes is untested and is a prerequisite for any operational claim.

Counterfactual scope. Evacuation dose–response establishes a stable intervention-response pattern within the learned model; it does not identify real-world causal effect sizes, and four of five intervention channels remain unreliable at demo scale (S9.3). Presenting only the evacuation channel without S9.3 would overstate the world model's maturity; both are therefore reported.

Data representativeness. Training is restricted to Atlantic tropical cyclones from 1995–2024. The vulnerability layer (CDC SVI 2022, Florida geography) embeds its own measurement choices and covers a single state, while the population and facility layers are synthetic (S1–S2), so exposure and prioritization outputs inherit both the SVI measurement assumptions and the assumptions of the synthetic generative protocol; errors in exposure or prioritization outputs may fall asymmetrically across populations — particularly where input data quality correlates with the vulnerability the system is meant to protect.

Dual-use and misuse. The system estimates, at a research-prototype level, where populations and facilities are most exposed. Outputs are scenario simulations with quantified miscalibration at long horizons (S8) and must not be used as a sole basis for resource allocation, evacuation ordering, or triage. Any deployment pathway requires validation against observed damage records, multi-basin evaluation, human-in-the-loop review by emergency-management practitioners, and uncertainty-aware decision procedures.